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BIOLOGY

Paper 3

Practicals**Time: 2 Hours****Marking scheme.*****Kenya Certificate of Secondary Education (K.C.S.E)***

1. a) Kingdom plantae. (1mk)
(Rej.wrong spelling)

Reasons;

- Contains photosynthetic pigments i.e. chlorophyll.
- Indefinite growth.
- Store lipids in form of oils.
- Store carbohydrates in form of starch.
- Contains cell wall.
- Autotrophic.

(2mks)

- b) (i)Phylum Arthropoda. (rej wrong spelling)

(ii)J-class Arachnida. (1mk)

F-class insect (1mk)

(rej. Wrong spelling)

(iii)-2 body parts.

-Four pairs of legs/eight walking leg

-Cephalothoraxes and abdomen.

-Simple eyes.

(Max 3mks)

- c) (i)Angiospermatophyta. (1mk)

(ii)

Dicotyledonous	monocotyledonous
-Network venation	-Parallel venation
-Has petiole	-Has sheath-like petiole
-narrow leaf	-Broad leaf

(Max 2mks)

2.

- a) X.....Humerus✓1
Y.....Radius✓1
Z.....Ulna✓1 (rej spellings)
- b) (i) A and C.....Analogous structures. ✓1
(ii) B and C.....Homologous structures✓1

c) Reasons for b (i) and (ii) above.

B and C-have common embryonic origin but have been modified to perform different functions.✓1

A and C-Have different embryonic origin but have been modified to perform same functions.✓ 1

d) A and C-convergent evolutions✓1

B and C-Divergent evolution✓1

e) -Wings of A originate from an exoskeleton while those of C originate from an endoskeleton✓1



-Wing of A has no pentadacty/structure while the wing of C has pentadacty/structure✓ 1

f) Organs that are reduced in size and function in organism✓1

3.

Food substance	Procedure	Observation	conclusion
Starch	-To solution W add (2dropsof)iodine solution✓1	-Brown/yellow color persists/remains✓1 aCc.no color change observed.	-starch absent✓1
Reducing sugars	-to solution W, add Benedict's solution and heat✓1/boil/warm.	-color/brown changes from blue to green to yellow to orange✓1 Acc.final color. Rej-red color	-Reducing sugars present✓1
Non-reducing sugars	-To solution W add dilute hydrochloric acid, heat and cool. Add sodium hydrogen carbonate solution until fizzing stops, then add Benedict's solution and heat/boil/warm✓1	-Color changes from blue to green to yellow to orange✓1/brown. Acc.final color. Rej.red color	Non-reducing sugars present✓1
Protein	-To solution W add sodium hydroxide then copper (ii) sulphate✓1 -Rej. Heat.	-Color changes from blue to purple✓1(purple color observed)	-Protein present✓1

(12mks)